Tactix no-MAB milestone and online-learning readiness

Codex for Andrew

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Decision. The user accepted the no-MAB stabilization milestone on 28 September 2026 after reviewing the four fixed-set results, the terminal Opus fuzzing report, and the remaining D2F proof-reach question. This is an explicit project checkpoint, not a claim that every N1–N4 universal requirement has been proved. Tactix source `d170b4a5` and the pinned Release binary SHA-256 beginning `2fc7421f` contain no MAB/CHB implementation. No source rule, legal public X7 action, or canonical checker contract was narrowed or changed for this acceptance. The local commits have not been pushed, as directed.

Observed verification. Rank 1–50 solved 50/50 (25 SAT, 25 UNSAT) in 7,582.072 solver seconds, inside the user's 5% leeway. Forty-nine cases completed the current independent automatic-proof/model route. Rank 22 solved UNSAT and emitted exactly the same 790,326,542-byte certificate previously verified via DRAT/CakeLPR, but the required no-option `dsr-trim` stage timed out inside its 16,000-second window; a same-binary, 45,000-second canonical diagnostic is running as `1734775_0`. B28+C5 finished 33/33 independently checked in 968.588 seconds, below the 999.876-second ceiling. WV100 finished 100/100 independently checked (33 SAT models, 67 UNSAT proofs) in 2,750.012 seconds. WV2–89 finished 79/89 checked; ten solver runs timed out at 600 seconds. The user accepted this four-set status as the no-MAB checkpoint. Pinned full Debug `1732906` passed all 188 tests.

Fuzzing and proof limits. Claude Code Opus 5.5 ran the frozen AM531 no-MAB fuzz campaign: 13,800 public C-API programs under seven writers and 9,500 CLI cases. It found zero solver crashes, invalid models, rejected produced certificates, route mismatches, or parity failures. The frozen campaign itself is *not* marked passed: nine truncation negative controls were inapplicable because their kept prefixes already refuted the CNF; an uncounted correction was rejected 9/9. The public D2F critical intersection was not reached. Exact older direct-Proof-API reproducers require proof-only variables and direct deletions that are not legal public X7 actions on their original CNFs; no proof excludes an analogous legal public sequence. This is a recorded residual risk for any future learning-driven public action policy, not a newly observed public solver failure.

Tier D experiment. The four competition-winner arms (Kissat-SC2024, LymphoSAT, AE-Kissat-MAB, Satsuma+Kissat) are running as array `1733043` on `cpu-q`; each task has four CPUs, 32 GiB, 5,000 seconds for solve and 14,400 seconds for its own independent proof checker. The common array is currently throttled to 119 while the single Rank diagnostic runs, keeping at most 120 benchmark tasks active. The full 1,166-instance no-MAB Tactix array `1735296` has been submitted with the same resources and a 120-task throttle; it starts only after `1733043` terminates. Its compute-node model, proof-route and abnormal-cleanup validation passed before submission. Anonymous node-local scratch frees decompressed CNFs and proofs on termination. Ten-minute

repository-storage guards hold only pending tasks above 700 GB; a dependent guard pair is queued for the Tactix array. Full Tier D performance and N3 parity remain unknown until terminal results are checked.

Online-learning readiness. The code already exposes ordered, typed inprocessing steps, instance parameters, repeat and work budgets, and a configurable DefaultPolicy schedule. Branching and clause-deletion policy interfaces allow pre-solve substitution, but safe live switching still lacks policy-state ownership, transition boundaries, state migration, and reward attribution; the reduction interface currently chooses a prefix after existing ranking rather than a fully independent retention policy. Therefore this codebase is *not ready* for a learner that switches inprocessing, branching and deletion strategies during a solve. A next session should design those contracts, then verify legal X7 reach and independent proof/model checks for learned action sequences, and score the terminal Tier D comparison before making a competitive-default claim. Reintroducing MAB is not required for that work.

Evidence. `benchmarks/runs/20260928-am531-no-mab-opus/REPORT.md`; `benchmarks/reports/2026-09-28-am542-rank22-canonical-checker-timeout.md`; `benchmarks/reports/2026-09-28-am533-learning-readiness-code-audit.md`; `benchmarks/reports/2026-09-28-am540-four-winner-tierd-cpuq-launched.md`; `benchmarks/runs/20260928-am541-no-mab-tactix-tierd-cpuq/launch.json`; `PROGRESS.md`.